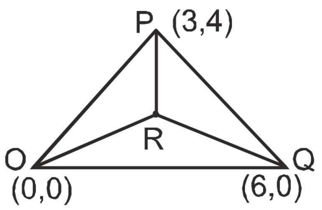
Exercise J01

Q1. Let O(0, 0), P (3, 4), Q(6, 0) be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are

(A) (4/3,3) (B) (3, 2/3) (C) (3 , 4/3) (D) (4/3 , 2/3)

Ans. C

Sol.



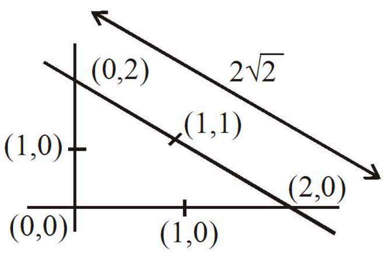
R is centroid hence R $\equiv \left(3, \frac{4}{3}\right)$

Q2. The x–coordinate of the incentre of the triangle that has the coordinates of mid point of its sides as (0, 1) (1, 1) and (1, 0) is :

(A) $2+\sqrt{2}$ (B) $2-\sqrt{2}$ (C) $1+\sqrt{2}$ (D) $1-\sqrt{2}$

Ans. B

Sol.



$$x = \frac{ax_1 + bx_2 + cx_3}{a+b+c}$$

$$= \frac{2(0) + 2\sqrt{2}(0) + 2(2)}{2+2+2\sqrt{2}}$$

$$= \frac{4}{4+2\sqrt{2}} = \frac{2}{2+\sqrt{2}} = 2-\sqrt{2}$$

Q3. Let the orthocentre and centroid of a triangle be A(–3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is

(A) $3\sqrt{\frac{5}{2}}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\sqrt{10}$ (D) $2\sqrt{10}$

Ans. A

Sol.

B divides AC in 2 : 1

$AC = \frac{3AB}{2}$; $\qquad r = \frac{AC}{2} = \frac{3AB}{4}$

$= \frac{3\sqrt{40}}{4}$; $\qquad = 3\sqrt{\left(\frac{5}{2}\right)}$

Q4. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points $(x_1\ y_1)$, (x_2, y_2) and (x_3, y_3)

(A) are vertices of a triangle (B) lie on a straight line (C) lie on an ellipse

(D) lie on a circle

Ans. B

Sol. Here a + b = 1. Required line is

$\frac{x}{a} - \frac{y}{1+a} = 1$ ..... (i)

Since line (i) passes through (4, 3)

$\therefore \frac{4}{a} - \frac{3}{1+a} = 1 \Rightarrow 4 + 4a – 3a = a + a^2$

$\Rightarrow a^2 = 4 \Rightarrow a = 2$

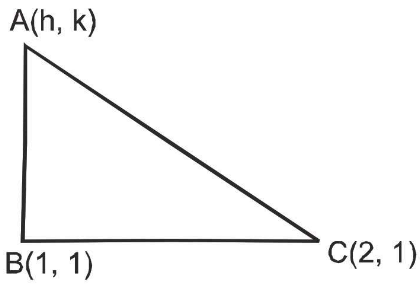
$\therefore$ Required lines are $\frac{x}{2} - \frac{y}{3} = 1$ & $\frac{x}{-2} - \frac{y}{1} = 1$

Q5. Let A (h, k),B(1, 1) and C (2, 1) be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which 'k' can take is given by

(A) $\{–1, 3\}$ (B) $\{–3, – 2)$ (C) $\{1, 3\}$ (D) $\{0, 2\}$

Ans. A

Sol.



AB = $\sqrt{(h-1)^2 + (k-1)^2}$
BC = 1

$\text{AC} = \sqrt{(h-2)^2 + (k-1)^2}$

$AB^2 + BC^2 = AC^2 \Rightarrow (h-1)^2 + (k-1)^2 + 1 = (h-2)^2 + (k-1)^2 \Rightarrow 2h = 2 \Rightarrow h = 1$

$\triangle ABC = \frac{1}{2}\sqrt{(h-1)^2 + (k-1)^2} \times 1 = 1$

$(K-1)^2 = 4 \Rightarrow k - 1 = \pm 2 \Rightarrow k = 3, -1.$

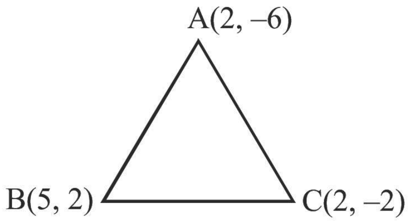
Q6. Let k be an integer such that the triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point.

(A) $\left(1, -\frac{3}{4}\right)$ (B) $\left(2, \frac{1}{2}\right)$ (C) $\left(2, -\frac{1}{2}\right)$ (D) $\left(1, \frac{3}{4}\right)$

Ans. B

Sol. $\begin{vmatrix} k & -3k & 1 \\ 5 & k & 1 \\ -k & 2 & 1 \end{vmatrix} = \pm 56$

$\Rightarrow$ k = 2 (k is an integer)



orthocenter $\equiv \left(2, \frac{1}{2}\right)$

Q7. A square of side a lies above the x–axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x–axis. The equation of its diagonal not passing through the origin is:

(A) y (cos α + sin α) + x (cos α – sin α) = a

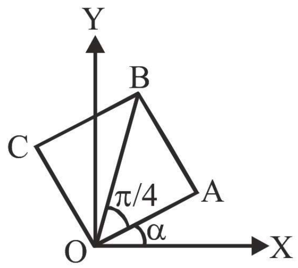
(B) y (cos α – sin α) – x (sin α – cos α) = a

(C) y (cos α + sin α) + x (sin α – cos α) = a

(D) y (cos α + sin α) + x (sin α + cos α) = a

Ans. A

Sol. Co-ordinates of A = (acosα, asinα)



Equation of OB, y = tan $\left(\frac{\pi}{4}+\alpha\right)x$

$\because$ CA $\perp^{r}$ to OB

$\therefore$ Slope of CA = –cot $\left(\frac{\pi}{4}+\alpha\right)$

Equation of CA,

y – asina = –cot $\left(\frac{\pi}{4}+\alpha\right)$ (x – acosα)

$\Rightarrow$ y(sinα + cosα) + x(cosα – sinα) = a
$\Rightarrow$ y(cosα + sinα) – x(sina – cosα) = a

Q8. The equation of the straight line passing through the point (4, 3) and making intercepts on the co–ordinate axes whose sum is –1 is:

(A) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$ (B) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$
(C) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{2}+\frac{y}{1}=1$ (D) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

Ans. A

Sol. -

Q9. The line parallel to the x–axis and passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0, where (a, b) $\neq$ (0, 0) is:

(A) below the x–axis at a distance of $\frac{3}{2}$ from it

(B) below the x–axis at a distance of $\frac{2}{3}$ from it

(C) above the x–axis at distance of $\frac{3}{2}$ from it

(D) above the x–axis at a distance of $\frac{2}{3}$ from it

Ans. A

Sol. The lines passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0 is
ax + 2by + 3b + λ(bx – 2ay – 3a) = 0
$\Rightarrow$ (a + bλ)x + (2b – 2aλ)y + 3b – 3λa = 0 ..... (i)
Line (i) is parallel to x-axis

$\therefore a + b\lambda = 0 \Rightarrow \lambda = \frac{-a}{b} = 0$

Put the value of λ in (i)

$ax + 2by + 3b - \frac{a}{b}(bx - 2ay - 3a) = 0$

$y\left(2b + \frac{2a^2}{b}\right) + 3b + \frac{3a^2}{b} = 0,$

$y\left(\frac{2b^2 + 2a^2}{b}\right) = -\left(\frac{3b^2 + 3a^2}{b}\right)$

$y = \frac{-3(a^2 + b^2}{2(b^2 = a^2)} = \frac{-3}{2}, y = -\frac{3}{2}$

So, it is $\frac{3}{2}$ unit below x-axis

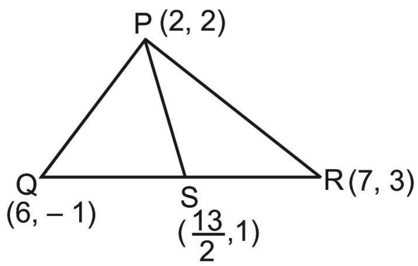
Q10. Let PS be the median of the triagle with vertices P(2, 2), Q(6, –1) and R(7, 3). The equation of the line passing through (1, –1) and parallel to PS is:

(A) 4x + 7y + 3 = 0 (B) 2x – 9y – 11 = 0 (C) 4x – 7y – 11 = 0

(D) 2x + 9y + 7 = 0

Ans. D

Sol.



Slope of PS = $\frac{2-1}{2-\frac{13}{2}} = \frac{-2}{9}$

Hence equation of line through (1, –1) & parallel to PS is:

$(y+1) = \frac{-2}{9}(x-1)$

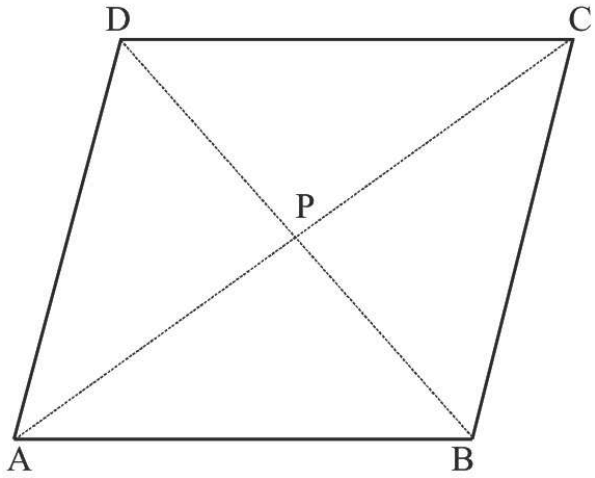
9y + 2x + 7 = 0

Q11. Two sides of a rhombus are along the lines, x – y + 1 = 0 and 7x – y – 5 = 0. If its diagonals intersect at (–1, –2), then which one of the following is a vertex of this rhombus ?

(A) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (B) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (C) (–3, –9) (D) (–3, –8)

Ans. A

Sol.



AB = 7x – y – 5 = 0

AD = x – y + 1 = 0

A(+1, 2)

P(–1, – 2)

⇒ C(–3, –6)

BC = x – y – 3 = 0

CD = 7x – y + 15 = 0

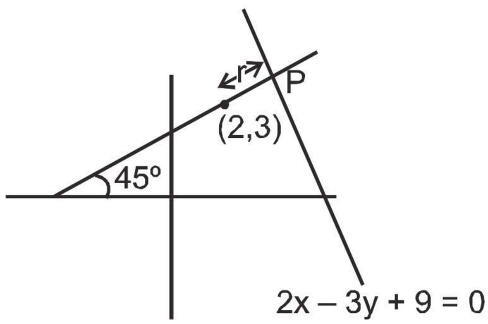
$\Rightarrow B\left(\frac{1}{3}, \frac{-8}{3}\right)$

Q12. The distance of the point (2, 3) from the line 2 x – 3 y + 9 = 0 measured along a line x – y + 1 = 0 is :

(A) $5\sqrt{3}$ (B) $4\sqrt{2}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

Ans. B

Sol. Let coordinates of point P by parametric P(2 + r cos 45º, 3 + r sin 45º)



It satisfies the line 2x – 3y + 9 = 0 ; $2\left(2+\frac{r}{\sqrt{2}}\right)-3\left(3+\frac{r}{\sqrt{2}}\right)+9=0 \Rightarrow r=4\sqrt{2}$

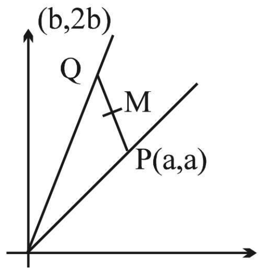
Q13. P lies on the line y = x and Q lies on y = 2x. The equation for the locus of the mid point of PQ, if | PQ | = 4, is

(A) $25x^2 + 36xy + 13y^2 = 4$ (B) $25x^2 – 36xy + 13y^2 = 4$

(C) $25x^2 - 36xy - 13y^2 = 4$ (D) $25x^2 + 36xy - 13y^2 = 4$

Ans. B

Sol.



$2h = a + b$

$2k = a + 2b$

$\therefore \quad b = 2(k - h)$

$a = 2(2h - k)$ now proceed.

Q14. The line L_1 given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point M(13, 32). The line L_2 is parallel to L_1 and has the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L_1 and L_2 is

(A) $\sqrt{17}$ (B) $\frac{17}{\sqrt{15}}$ (C) $\frac{23}{\sqrt{17}}$ (D) $\frac{23}{\sqrt{15}}$

Ans. C

Sol. Slope of line $L_1 = \frac{-b}{5}$ and slope of line $L2 = \frac{-3}{c}$

Line L_1 is parallel to line $L_2 \Rightarrow \frac{b}{5} = \frac{3}{c} \Rightarrow bc = 15$

As (13, 32) lies on L_1, so $\frac{13}{5} + \frac{32}{b} = 1 \Rightarrow \frac{32}{b} = \frac{-8}{5} \Rightarrow b = -20$

So, $c = \frac{-3}{4}$

$\therefore$ Equation of $L_2 : y - 4x = 3$

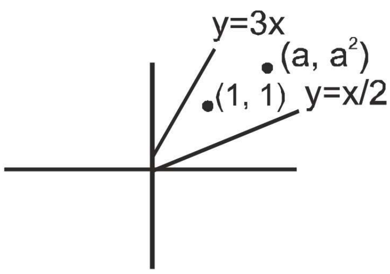
Hence distance between L_1 and $L_2 = \frac{|52 - 32 + 3|}{\sqrt{17}} = \frac{23}{\sqrt{17}} = .$ **Ans.**

Q15. If (a, a^2) falls inside the angled made by the lines $y = \frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then a along to

(A) $\left(0, \frac{1}{2}\right)$ (B) $(3, \infty)$ (C) $\left(\frac{1}{2}, 3\right)$ (D) $\left(-3, -\frac{1}{2}\right)$

Ans. C

Sol.



Let $L_1(x - 2y)$ & $L_2(3x - y)$

Since (1, 1) and (a, a2) Both lies same side with respect to both lines

$L_1(1, 1) \times L_1(a, a^2) > 0 \quad \Rightarrow \quad a - 2a^2 < 0 \;\Rightarrow\; 2a^2 - a > 0$

$\Rightarrow a(2a - 1) > 0$

similarly $L_2(1, 1) \times L2(a, a2) > 0 \;\Rightarrow$Þ $3a - a2 > 0 \Rightarrow a^2 - 3a < 0 \Rightarrow a \in (0, 3)$

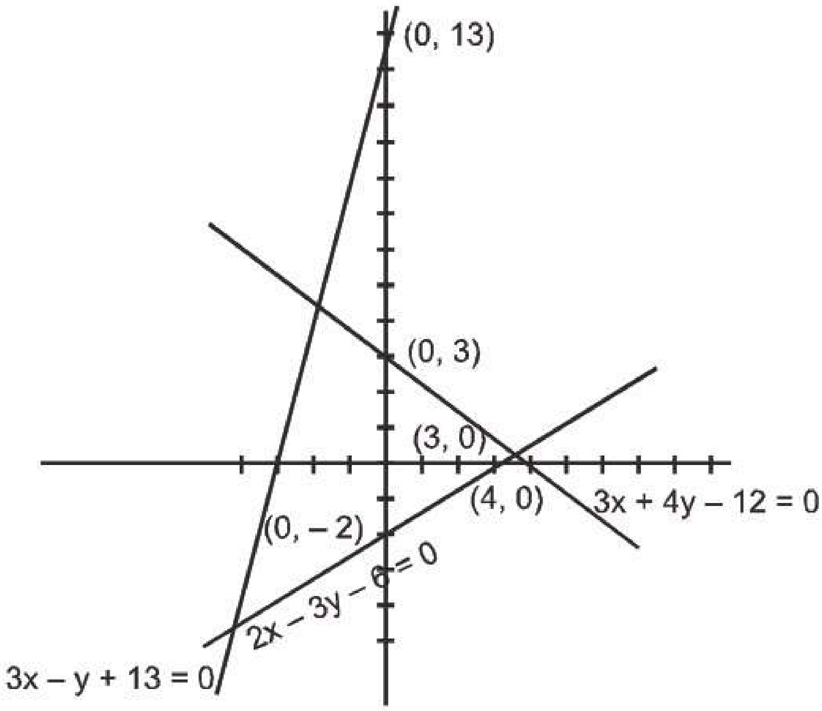
Q16. A triangle ABC is formed by the lines $2x - 3y - 6 = 0$; $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha, 0)$ and $Q(0, \beta)$ always lie on or inside the Δ ABC, then :

(A) $\alpha \in [-1, 2]$ and $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ and $\beta \in [-2, 4]$

(C) $\alpha \in [-2, 4]$ and $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ and $\beta \in [-2, 3]$

Ans. D

Sol. $L_1 = 2x - 3y - 6 = 0$



$L_2 = 3x - y + 3 = 0 \quad \Rightarrow \quad L_3 = 3x + 4y - 12 = 0$

$\alpha \in [-1, 3] \;\Rightarrow\; \beta \in [-2, 3]$

Q17. If the lines

$$\left.\begin{aligned} \lambda x + (\sin\alpha)y + \cos\alpha = 0 \\ x + (\cos\alpha)y + \sin\alpha = 0 \\ x + (\sin\alpha)y + \cos\alpha = 0 \end{aligned}\right]$$ pass through the same point where a $\in$ R then λ lies in the interval

(A) $[-1, 1]$ (B) $[-\sqrt{2}, \sqrt{2}]$ (C) $[-2, 2]$ (D) $(-\infty, \infty)$

Ans. B

Sol. -

Q18. The lines ax + by + c = 0, where 3a + 2b + 4c = 0 are concurrent at the point :

(A) $\left(\frac{1}{2}, \frac{3}{4}\right)$ (B) $(1, 3)$ (C) $(3, 1)$ (D) $\left(\frac{3}{4}, \frac{1}{4}\right)$

Ans. D

Sol. ax + by + c = 0 $\Rightarrow$ $\frac{3a}{4} + \frac{b}{2} + c = 0$

compare both $(x, y) \equiv \left(\frac{3}{4}, \frac{1}{2}\right)$. Hence given family passes through $\left(\frac{3}{4}, \frac{1}{2}\right)$.

Q19. Given the family of lines, a (2x + y + 4) + b (x – 2y – 3) = 0. Among the lines of the family, the number of $\sqrt{10}$lines situated at a distance of from the point M (2, – 3) is :

(A) 0 (B) 1 (C) 2 (D) ∞

Ans. B

Sol. $(2x + y + 4) + \lambda(x - 2y - 3) = 0$

$(2 + \lambda)x + 1(1 - 2\lambda)y + (4 - 3\lambda) = 0$

m(2, –3)

$$\frac{4 + 2\lambda + 6\lambda - 3 + 4 - 3\lambda}{\sqrt{(2 \times \lambda)^2 + (1 - 2\lambda)^2}} = \pm\sqrt{10}$$

$$\Rightarrow \frac{(\lambda + 1)^2}{(\lambda^2 + 1)} = 2$$

$\Rightarrow \lambda^2 + 1 + 2\lambda = 2\lambda^2 + 2$

$\Rightarrow \lambda^2 + 1 - 2\lambda = (\lambda - 1)^2 = 0$

$\lambda = 1$

Q20. The equation of image of pair of lines y = |x – 1| with respect to y–axis is

(A) $x^2 - y^2 - 2x + 1 = 0$ (B) $x^2 - y^2 - 4x + 4 = 0$ (C) $4x^2 - 4x - y^2 + 1 = 0$

(D) $x^2 - y^2 + 2x + 1 = 0$

Ans. D

Sol. -

Q21. A light beam emanating from the point A(3, 10) reflects from the straight line 2x + y – 6 = 0 and then passes through the point B(4, 3). The equation of the reflected beam is :

(A) $3x - y + 1 = 0$ (B) $x + 3y - 13 = 0$ (C) $3x + y - 15 = 0$ (D) $x - 3y + 5 = 0$

Ans. B

Sol.

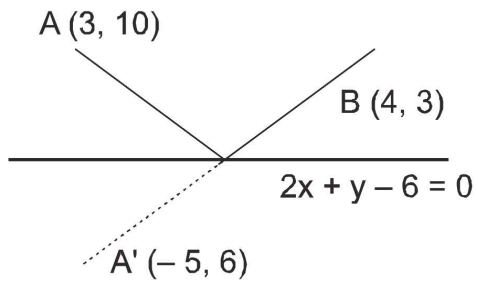


Image of A(3, 10) in $2x + y - 6 = 0$

$$\frac{x-3}{2} = \frac{y-10}{1} = -2\left(\frac{6+10-6}{2^2+1^2}\right)$$

$$\frac{x-3}{2} = \frac{y-10}{1} = -4$$

Equation of A'B is

$$y - 3 = \left(\frac{6-3}{-5-4}\right)(x-4) \quad \Rightarrow \quad y - 3 = -\frac{1}{3}(x-4)$$

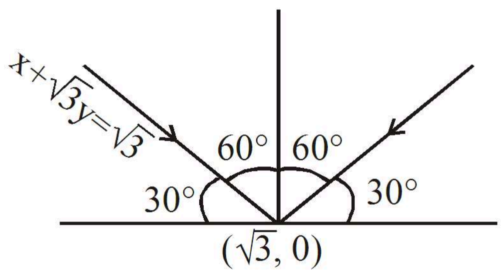
$3y - 9 = -x + 4 \quad \Rightarrow \quad x + 3y - 13 = 0$

Q22. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x–axis, the equation of the reflected ray is :

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$ (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

Ans. B

Sol.



$$y = \frac{1}{\sqrt{3}}(x - \sqrt{3})$$

$$\sqrt{3}y = x - \sqrt{3}$$

Q23. Find the equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$.

(A) $11x - 3y + 9 = 0$ (B) $3x + 11y - 13 = 0$ (C) $3x + 11y - 3 = 0$ (D) $11x - 3y + 2 = 0$

Ans. A

Sol. Equation of bisectors are

$$\left(\frac{3x-4y+7}{5}\right)=\pm\left(\frac{-12x-15y+2}{13}\right)$$

Here $a_1a_2 + b_1b_2 = -36 + 20 = -20$

Hence acute angle bisector is

$$\left(\frac{3x-4y+7}{5}\right)=\left(\frac{2-12x-5y}{13}\right)$$

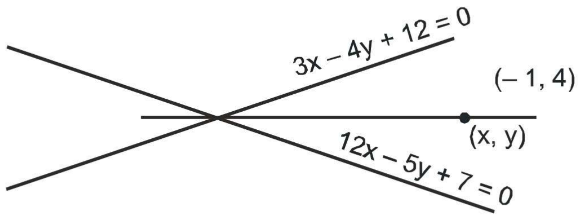
Q24. The equation of the bisector of the angle between two lines $3x - 4y + 12 = 0$ and $12x - 5y + 7 = 0$ which contains the points $(-1, 4)$ is :

(A) $21x + 27y - 121 = 0$ (B) $21x - 27y + 121 = 0$ (C) $21x + 27y + 191 = 0$

(D) $\frac{-3x+4y-12}{5}=\frac{12x-5y+7}{13}$

Ans. A

Sol.



at $(-1, 4)$

$3x - 4y + 12 < 0$ and $12x - 5y + 7 < 0$

$\Rightarrow \frac{3x-4y+12}{12x-5y+12} > 0$ at $(-1, 4)$

So we have to take the bisector with + sign

$$\frac{3x-4y+12}{5}=\frac{12x-5y+7}{13} \Rightarrow 21x + 27y - 121 = 0$$

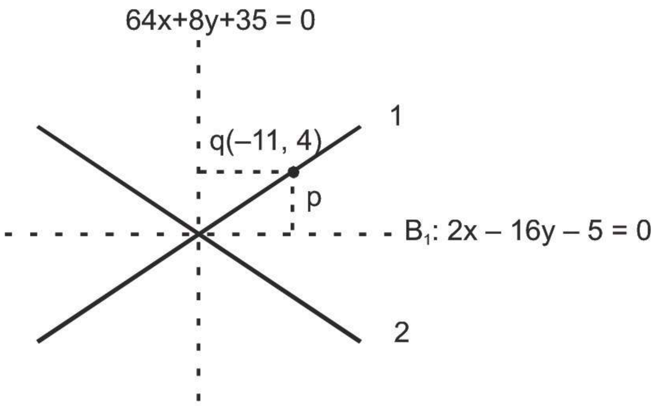
Q25. The equation of bisectors of two lines L_1 & L_2 are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line L_1 passes through $(-11, 4)$, the equation of acute angle bisector of L_1 & L_2 is :

(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) data insufficient

(D) none of these

Ans. A

Sol.



$$p = \left| \frac{-22 - 64 - 5}{\sqrt{2^2 + (-16)^2}} \right| = \frac{91}{\sqrt{260}}$$

$$q = \left| \frac{-64\times 11 + 8\times 4 + 35}{\sqrt{64^2 + (-16)^2}} \right| = \frac{637}{\sqrt{4160}} = \frac{159.25}{\sqrt{260}}$$

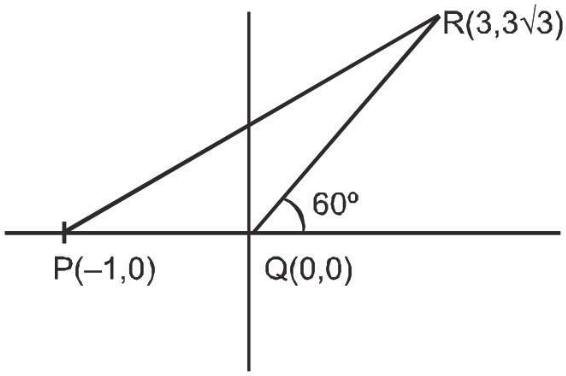
p < q Hence 2x – 16y – 5 = a cute angle besectory

Q26. Let P = (–1, 0), Q = (0, 0) and R $(3, 3\sqrt{3})$ be three point. The equation of the bisector of the angle PQR is:

(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3}y = 0$ (C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

Ans. C

Sol.



The line segment QR makes an angle 60º with the positive direction of x-axis.
hence bisector of angle PQR will make 120º with +ve direction of x-axis.
∴ Its equation
y – 0 = tan120º (x – 0) ⇒ y = – $\sqrt{3}$ x ⇒ x$\sqrt{3}$ + y = 0

Q27. The co–ordinates of a point P on the line 2x – y + 5 = 0 such that |PA – PB| is maximum where A is (4, – 2) and B is (2, – 4) will be :

(A) (11, 27) (B) (– 11, – 17) (C) (– 11, 17) (D) (0, 5)

Ans. B

Sol. -

Q28. Locus of mid point of the portion between the axes of x cos α + y sin α = p whre p is constant is

(A) $x^2+y^2=\frac{4}{p^2}$ (B) $x^2+y^2=4p^2$ (C) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{2}{p^2}$ (D) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

Ans. D

Sol. -

Q29. Locus of centroid of the triangle whose vertices are (a cos t, a sin t), (b sin t, – b cos t) and (1, 0) where t is a parameter, is:

(A) $(3x+1)^2+(3y)^2=a^2-b^2$ (B) $(3x-1)^2+(3y)^2=a^2-b^2$

(C) $(3x-1)^2+(3y)^2=a^2+b^2$ (D) $(3x+1)^2+(3y)^2=a^2+b^2$

Ans. C

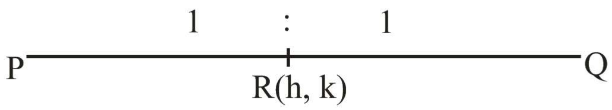
Sol. -

Q30. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is

(A) $y^2-4x+2=0$ (B) $y^2+4x+2=0$ (C) $x^2+4y+2=0$ (D) $x^2-4y+2=0$

Ans. A

Sol. $O\left(\frac{t^2}{8},t\right)$ P(1, 0)



$h=\frac{\frac{y^2}{8}+1}{2}$ $k=\frac{t}{2}$

$(2h-1)8=t^2$ ...... (1) $t=2k$ ...... (2)

from (1) & (2)

$(2h-1)^8=4k^2$

$y^2=2(2x-1)$

Q31. The equatons $x = t^3 + 9$ and $y=\frac{3t^3}{4}$ represents a straight line where t is a parameter. Then y–intercept of the line is

(A) $-\frac{3}{4}$ (B) 9 (C) 6 (D) 1

Ans. A

Sol. -

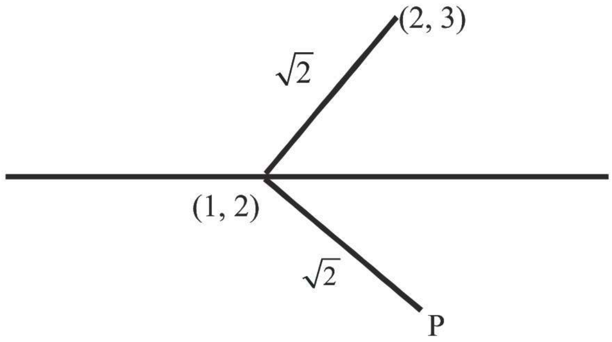
Q32. Locus of the image of the point (2, 3) in the line (2x – 3y + 4) + k (x – 2y + 3) = 0 k Î R is a :

(A) straight line parallel to x–axis (B) straight line parallel to y–axis

(C) circle of radius $\sqrt{2}$ (D) circle of radius $\sqrt{2}$

Ans. C

Sol.



$2x - 3y + 4 = 0$

$x - 2y + 3 = 0$

Line passing through (1, 2)

P will at distance of $\sqrt{2}$ from (1, 2)

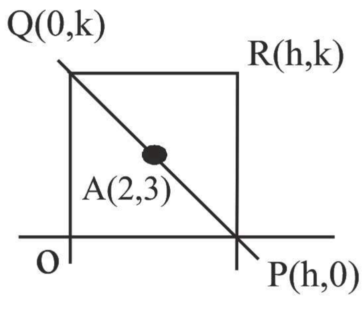
So locus of P will circle of radius $\sqrt{2}$

Q33. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is

(A) $3x + 2y = xy$ (B) $3x + 2y = 6xy$ (C) $3x + 2y = 6$ (D) $2x + 3y = xy$

Ans. A

Sol.



P, A, Q will be collinear

$$\begin{vmatrix} h & 0 & 1 \\ 2 & 3 & 1 \\ 0 & k & 1 \end{vmatrix} = 0$$

$3h + 2k = hk$

locus of R (h, k) will be

$3x + 2y = xy$

Q34. If the slope of one line of the pair of lines represented by $ax^2 + 10xy + y^2 = 0$ is four times, the slope of the other line, then a =

(A) 1 (B) 2 (C) 4 (D) 16

Ans. D

Sol. $m_1 + m_2 = -10 \quad \Rightarrow \quad m_1 m_2 = \frac{a}{1}$

given $m_1 = 4m_2 \quad \Rightarrow \quad m_2 = -2,\ m_1 = -8,\ a = 16$

Q35. Tha pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)\,y^2 = 0$ are perpendicular to each other for

(A) two values of a (B) $\forall\, a \in R$ (C) for one value of a (D) for no value of a

Ans. A

Sol. -

Q36. If one of the lines of $my^2 + (1 - m^2)\,xy - mx^2 = 0$ is a bisector of the angle between the lines xy = 0, then m is

(A) 1 (B) 2 (C) $-\frac{1}{2}$ (D) -2

Ans. A

Sol. Bisector of x = 0 and y = 0 is either y = x or y = –x

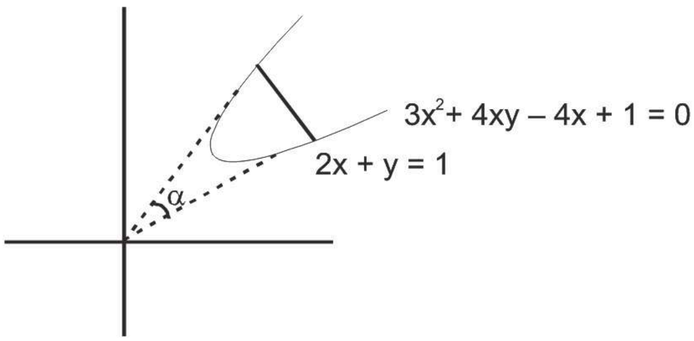
If y = x is Bisector, then

$mx^2 + (1 - m^2)x^2 - mx^2 = 0 \Rightarrow m + 1 - m^2 - m = 0 \Rightarrow m^2 = 1 \Rightarrow m = \pm 1.$

Q37. The straight lines joining the origin to the points of intersection of the line $2x + y = 1$ and curve $3x^2 + 4xy - 4x + 1 = 0$ include an angle :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Ans. A

Sol. 

Homogenize given curve with given line

$3x^2 + 4xy - 4x(2x + y) + 1(2x + y)^2 = 0$

$3x^2 + 4xy - 8x^2 - 4xy + 4x^2 + y^2 + 4xy = 0$

$-x^2 + 4xy + y^2 =$

coeff. x^2 + coeff. $y^2 = 0$

Hence angle is 90º

Q38. If the straight lines joining the origin and the points of intersection of the curve

$$5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0 \text{ and } x + ky - 1 = 0$$

are equally inclined to the co–ordinate axes then the value of k :

(A) is equal to 1 (B) is equal to -1 (C) is equal to 2

(D) does not exist in the set of real numbers .

Ans. B

Sol. Homogenise and put co-efficient of $xy = 0$

Q39. The angles between the straight lines joining the origin to the points common to $7x^2 + 8y^2 - 4xy + 2x - 4y - 8 = 0$ and $3x - y = 2$ is :

(A) $\tan^{-1}\sqrt{2}$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/2$

Ans. D

Sol. -

Exercise J02

Q1. Determine the ratio in which the point P(3 , 5) divides the join of A(1, 3) & B(7, 9). Find the harmonic conjugate of P w.r.t. A & B.

Ans. 1 : 2 ; Q (–5, –3)

Sol. -

Q2. Line $\frac{x}{6} + \frac{y}{8} = 1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find

(a) the coordinates of the point P and

(b) the radius of the circle escribed opposite to the angle N.

Ans. (a) $\left(2, \frac{8}{3}\right)$; (b) 4

Sol. -

Q3. In an acute triangle ABC, the base BC has the equation 4x – 3y + 3 = 0. If the coordinates of the orthocentre (H) and circumcentre (P) of the triangle are (1, 2) and (2, 3) respectively, then the radius of the circle circumscribing the triangle is $\frac{\sqrt{m}}{n}$ where m and n are relatively prime. Find the value of (m + n).

(You may use the fact that the distance between orthocentre and circumcentre of the triangle is

given by R = $\sqrt{1 - 8\cos A\cos B\cos C}$).

Ans. 63

Sol. -

Q4. The point A divides the join of P (–5 , 1) & Q (3, 5) in the ratio K : 1 . Find the two values of K for which the area of triangle ABC, where B is (1, 5) & C is (7, –2), is equal to 2 units in magnitude.

Ans. K = 7 or $\frac{31}{9}$

Sol. -

Q5. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$ $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ and $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$ then

$\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a + b + c)(b + c - a)(c + a - b)(a + b - c)$. Find the value of λ.

Ans. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

Sol. -

Q6. The area of a triangle is 5. Two of its vertices are (2, 1) & (3, –2). The third vertex lies on y = x + 3. Find the third vertex.

Ans. $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$7

Sol. -

Q7. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle.

Ans. 3 units

Sol. -

Q8. The points (–6, 1), (6, 10), (9, 6) and (–3, –3) are the vertices of a rectangle. If the area of the portion of this rectangle that lies above the x axis is , find the value of (a + b), given a and b are coprime.

Ans. 533

Sol. -

Q9. Two vertices of a triangle are (4, –3) & (–2, 5). If the orthocentre of the triangle is at (1, 2), find the coordinates of the third vertex.

Ans. (33, 26)

Sol. -

Q10. A line is such that its segment between the straight lines 5x – y – 4 = 0 and 3x + 4y – 4 = 0 is bisected at the point (1, 5). Obtain the equation.

Ans. 83x – 35y + 92 = 0

Sol. oordinate of A which is at a distance 'r' from

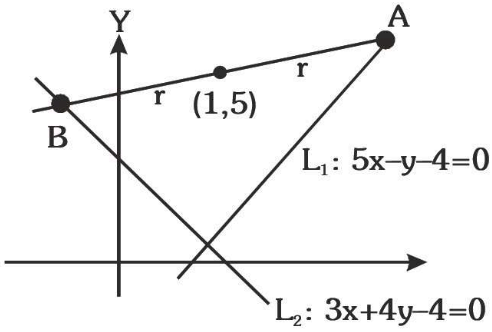
(1, 5) is $(1 + r\cos\theta, 5 + r\sin\theta)$

which satisfies $L_1 = 0$

$5(1 + r\cos\theta) - (5 + r\sin\theta) - 4 = 0$

$r(5\cos\theta - \sin\theta) = 4$

$$r = \frac{4}{5cos\theta - sin\theta}$$



Similarly coordinate of point B ($1 - r\cos\theta$, $5 - r\sin\theta$) which lies on $L_2 = 0$

$\therefore 3(1 - r\cos\theta) + 4(5 - r\sin\theta) - 4 = 0$

$$r = \frac{19}{3cos\theta + 4sin\theta}$$

Equating both values of r & get $\tan\theta$ (= m) and equation of the line passing through (1, 5).

Q11. Two consecutive sides of a parallelogram are $4x + 5y = 0$ & $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.

Ans. $x - y = 0$

Sol. -

Q12. The line $3x + 2y = 24$ meets the y–axis at A & the x–axis at B. The perpendicular bisector of AB meets the line through (0, –1) parallel to x–axis at C. Find the area of the triangle ABC.

Ans. 91 sq.units

Sol. -

Q13. Find the area of the triangle formed by the straight lines whose equations are $x + 2y - 5 = 0$;

$2x + y - 7 = 0$ and $x - y + 1 = 0$. Also compute the tangent of the interior angles of the triangle and hence comment upon the nature of triangle.

Ans. $\frac{3}{2}$sq. units, $\left(3, 3, \frac{3}{4}\right)$, isosceles

Sol. -

Q14. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

Ans. $c = -4$;   B(2 , 0);   D(4 , 4)

Sol. -

Q15. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

Ans. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

Sol. -

Q16. The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is $(1, -2)$ find the equation of the line BC.

Ans. $14x + 23y = 40$

Sol. -

Q17. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the Δ ABC are G, H and S respectively. $L = 0$ denotes the equation of side BC.

(a) If $L_1 : 2x - y = 0$ and $L_2 : x + y = 3$ and G (2, 3) then find the slope of the line $L = 0$.

(b)If $L_1 : 2x + y = 0$ and $L_2 : x - y + 2 = 0$ and H (2, 3) then find the y-intercept of $L = 0$.

(c)If $L_1 : x + y - 1 = 0$ and $L_2 : 2x - y + 4 = 0$ and S(2, 1) then find the x-intercept of the line $L = 0$.

Ans. (a) 5 ; (b) 2 ; (c) 3/2

Sol. -

Q18. Find the equation of the straight lines passing through $(-2, -7)$ & having an intercept of length 3 between the straight lines $4x + 3y = 12$, $4x + 3y = 3$.

Ans. $7x + 24y + 182 = 0$ or $x = -2$

Sol. -

Q19. Consider a D ABC whose sides AB, BC and CA are represented by the straight lines $2x + y = 0$, $x + py = q$ and $x - y = 3$ respectively. The point P is (2, 3).

(a) If P is the centroid, then find the value of $(p + q)$.

(b) If P is the orthocentre, then find the value of $(p + q)$.

(c) If P is the circumcentre, then find the value of $(p + q)$.

Ans. (a) 74 ; (b) 50; (c) 47

Sol. -

Q20. A line through the point $P(2, -3)$ meets the lines $x - 2y + 7 = 0$ and $x + 3y - 3 = 0$ at the points A and B respectively. If P divides AB externally in the ratio 3 : 2 then find the equation of the line AB.

Ans. $2x + y - 1 = 0$

Sol. -

Q21. If the straight line drawn through the point P $(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ.

Ans. 6 units

Sol. -

Q22. Consider the family of lines $(x - y - 6) + \lambda(2x + y + 3) = 0$ and $(x + 2y - 4) + \mu(3x - 2y - 4) = 0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

Ans. $x^2 + y^2 - 3x + 4y - 3 = 0$

Sol. -

Q23. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

Ans. 15

Sol. -

Q24. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point $(1, -10)$. Determine the equation of the third side.

Ans. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$

Sol. -

Q25. Given vertices A (1, 1), B (4, –2) & C (5, 5) of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

Ans. $x - 5 = 0$

Sol. -

Q26. A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a} + \frac{y}{b} = 1$ & $\frac{x}{b} + \frac{y}{a} = 1$, meets the coordinate axes in A & B. Find the locus of the mid point of AB.

Ans. $2xy(a + b) = ab(x + y)$

Sol. By family of lines $\left(\frac{x}{a} + \frac{y}{b} - 1\right) + \lambda\left(\frac{x}{b} + \frac{y}{a} - 1\right) = 0$ it cuts x-axis at $A\left(\frac{\lambda + 1}{\frac{1}{a} + \frac{\lambda}{b}}, 0\right)$

y - axis at $B\left(0, \frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}\right)$ Let mid point of AB is P(h, k) $\Rightarrow$ $2h = \frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}, 2k = \frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}$

eliminate λ from both equations we get $2hk(a+b) = ab(h+k)$

Hence locus of P(h, k) is $2xy(a+b) = ab(x+y)$.

Q27. In the xy plane, the line 'l_1' passes through the point (1, 1) and the line 'l_2' passes through the point (–1, 1). If the difference of the slopes of the lines is 2. Find the locus of the point of intersection of the lines l_1 and l_2

Ans. $y = x^2$ and $y = 2 - x^2$

Sol. $m_1 - m_2 = 2$ or $m_1 - m_2 = -2$

$\frac{k-1}{h-1} - \frac{k-1}{h+1} = 2$ $\qquad$ $\frac{k-1}{h-1} - \frac{k-1}{h+1} = -2$

$x^2 = y$ $\qquad$ $y = 2 - x^2$.

Q28. The area of the triangle formed by the intersection of a line parallel to x–axis and passing through P (h, k) with the lines y = x and x + y = 2 is $4h^2$. Find the locus of the point P.

Ans. $y = 2x + 1$, $y = -2x + 1$

Sol. A line passing through P(h, k) and parallel to x-axis is y = k. .....(i)

The other lines given are y = x .....(ii)

and $\qquad$ y + x = 2 .....(iii)

Let ABC be the D formed by the points of intersection of the lines (i), (ii) and (iii).

A(k, k) , B(1, 1), C (2–k, k)

$$\text{Area of } \triangle ABC = \frac{1}{2}\begin{vmatrix} k & k & 1 \\ 1 & 1 & 1 \\ 2-k & k & 1 \end{vmatrix} = 4h^2$$

$$C_1 \to C_1 - C_2 \quad \frac{1}{2}\begin{vmatrix} 0 & k & 1 \\ 0 & 1 & 1 \\ 2-2k & k & 1 \end{vmatrix} = 4h^2$$

$\Rightarrow \frac{1}{2}|(2-2k)(k-1)| = 4h^2 \quad \Rightarrow \quad (k-1)^2 = 4h^2$

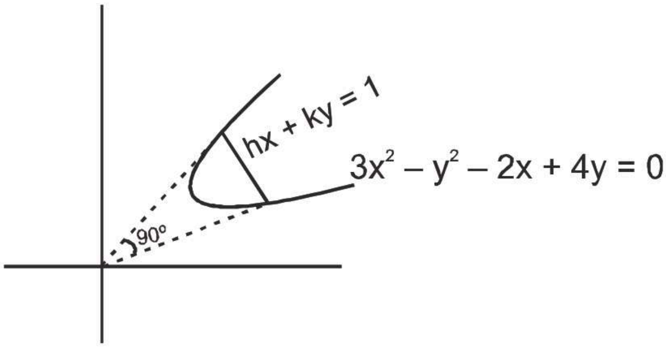
$\Rightarrow \quad k - 1 = 2h, \quad k - 1 = -2h \quad \Rightarrow \quad k = 2h + 1 \quad k = -2h + 1$

$\therefore$ locus of (h, k) is $y = 2x + 1 \quad y = -2x + 1$

Q29. Show that all the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrency & if not, give reasons.

Ans. $(1, 2)$, yes $\left(\frac{1}{3}, -\frac{2}{3}\right)$

Sol.



Let equation of chords $hx + ky = 1$

By homogenisation

$3x^2 - y^2 - 2x\,(hx + ky) + 4y\,(hx + ky) = 0$

$\therefore$ it makes 90º. Hence

coeff. x^2 + coeff. y2 = 0

$3 - 2h - 1 + 4k = 0 \Rightarrow h - 2k = 1$

Hence all chords are concurrent at $(1, -2)$

Q30. A straight line is drawn from the point $(1, 0)$ to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

Ans. $x + y = 1$; $x + 9y = 1$

Sol. -

Q31. If the straight line joining the origin to the points of intersection of $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ and $2x + 3y = k$ are at right angles, then find the value of $5k - 6k^2$.

Ans. 52

Sol. Curve : $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ .....(1)

Line : $2x + 3y = k \Rightarrow \frac{2x+.y}{k} = 1$ .....(2)

homogenise equation (1) with the help of equation (2), combined equation of line pair is

$$(3x^2 - xy + 3y^2) + (2x - 3y)\left(\frac{2x+3y}{k}\right) + 4\left(\frac{2x+3y}{k}\right)^2 = 0$$

given angle between the lines is 90° $\Rightarrow$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\left(3 + \frac{4}{k} + \frac{16}{k^2}\right) + \left(3 - \frac{9}{k} + \frac{66}{k^2}\right) = 0$$

$$6 - \frac{5}{k} + \frac{52}{k^2} = 0 \Rightarrow 6k^2 - 5k + 52 = 0$$

$\therefore$ $5k - 6k^2 = 52$ **Ans.**

Paragraph

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the

\npoints E(3, 4) and F(1, 2) respectively and centroid of D ABC is G(3, 2), then answer the following questions:

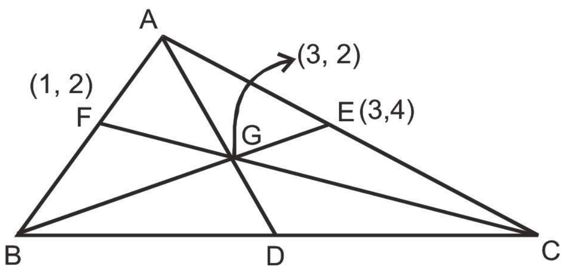
\n

Q32. The equation of side AB is

(A) $2x + y = 4$ (B) $x + y - 3 = 0$ (C) $4x - 2y = 0$ (D) none of these

Ans. A

Sol.



Let the co-ordinates of $D(\alpha, \beta)$ then $\frac{\alpha + 1 + 3}{3} = 3 \Rightarrow \alpha = 5$

$\therefore$ D(5, 0)

Taking $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

then by $\frac{x_1 + x_2}{2} = 1, \frac{x_2 + x_3}{2} = 5, \frac{x_3 + x_1}{2} = 3$ and $\frac{y_1 + y_2}{2} = 2, \frac{y_2 + y_3}{2} = 0, \frac{y_3 + y_3}{2} = 4$

we get A(–1, 6) , B(3, –2) , C(7, 2) equation of AB is 2x + y = 4

Height of altitude from A is = $\frac{2\times area(\triangle ABC)}{BC} = 6\sqrt{2}$

Paragraph

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the

\npoints E(3, 4) and F(1, 2) respectively and centroid of D ABC is G(3, 2), then answer the following questions:

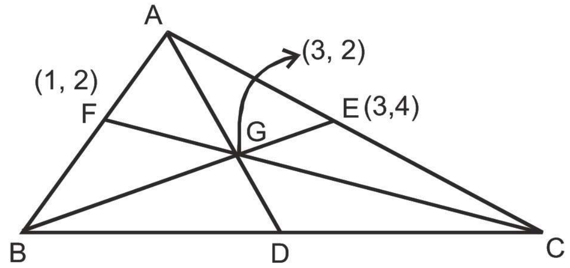
\n

Q33. Coordinate of D are

(A) $(7, -4)$ (B) $(5, 0)$ (C) $(7, 4)$ (D) $(-3, 0)$

Ans. B

Sol.



Let the co-ordinates of D(α, β) then $\frac{\alpha+1+3}{3} = 3 \quad \Rightarrow \quad \alpha = 5$

$\therefore$ D(5, 0)

Taking A(x_1, y_1) , B(x_2, y_2) and C(x_3, y_3)

then by $\frac{x_1+x_2}{2} = 1, \frac{x_2+x_3}{2} = 5, \frac{x_3+x_1}{2} = 3$ and $\frac{y_1+y_2}{2} = 2, \frac{y_2+y_3}{2} = 0, \frac{y_3+y_3}{2} = 4$

we get A(–1, 6) , B(3, –2) , C(7, 2) equation of AB is 2x + y = 4

Height of altitude from A is = $\frac{2\times area(\triangle ABC)}{BC} = 6\sqrt{2}$

Paragraph

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the

\npoints E(3, 4) and F(1, 2) respectively and centroid of D ABC is G(3, 2), then answer the following questions:

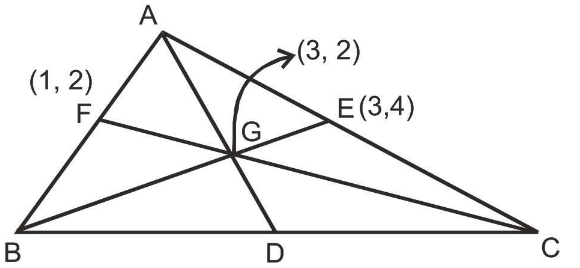
\n

Q34. Height of altitude drawn from point A is (in units)

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$ (C) $6\sqrt{2}$ (D) $2\sqrt{2}$

Ans. C

Sol.



Let the co-ordinates of D(α, β) then $\frac{\alpha+1+3}{3}=3 \quad \Rightarrow \quad \alpha=5$

$\therefore$ D(5, 0)

Taking A(x_1, y_1) , B(x_2, y_2) and C(x_3, y_3)

then by $\frac{x_1+x_2}{2}=1, \frac{x_2+x_3}{2}=5, \frac{x_3+x_1}{2}=3$ and $\frac{y_1+y_2}{2}=2, \frac{y_2+y_3}{2}=0, \frac{y_3+y_3}{2}=4$

we get A(–1, 6) , B(3, –2) , C(7, 2) equation of AB is 2x + y = 4

Height of altitude from A is = $\frac{2\times area(\triangle ABC)}{BC}=6\sqrt{2}$

Paragraph

Consider a triangle ABC with vertex A(2, –4). The internal bisectors of the angles B and C are x + y = 2 and
\nx – 3y = 6 respectively. Let the two bisectors meet at I.
\n

Q35. If (a, b) is incentre of the triangle ABC then (a + b) has the value equal to

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. -

Paragraph

Consider a triangle ABC with vertex A(2, –4). The internal bisectors of the angles B and C are x + y = 2 and
\nx – 3y = 6 respectively. Let the two bisectors meet at I.
\n

Q36. If (x_1, y_1) and (x_2, y_2) are the co–ordinates of the point B and C respectively, then the value of (x_1 x_2 + y_1 y_2) is equal to

(A) 4 (B) 5 (C) 6 (D) 8

Ans. D

Sol. -

Q37. The area of triangle ABC is 20 cm^2. The co-ordinates of vertex A are (- 5, 0) and B are (3, 0). The vertex C lies on the line, $x - y = 2$. The co-ordinates of C are :

(A) (5, 3) (B) (–3, –5) (C) (–5, –7) (D) (7, 5)

Ans. B, D

Sol. Let any point on the line x – y = 2 be C ≡ (h, h – 2). Given area of ΔABC is

$$\left|\frac{1}{2}\begin{vmatrix} h & h-2 & 1 \\ -5 & 0 & 1 \\ 3 & 0 & 1 \end{vmatrix}\right| = 20$$

$$\Rightarrow |8(h-2)| = 40$$

$$\Rightarrow h - 2 = \pm$$

$$\hbar = 7, -3$$

Hence, the points are (7,5) and (–3,–5)

Q38. Three vertices of a triangle are A(4, 3) ; B(1, - 1) and C(7, k) . Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the ΔABC lie on the same straight line is/are :

(A) 7 (B) - 1 (C) –9/8 (D) None

Ans. B, C

Sol. AB = AC

$\Rightarrow 9 + 16 = 9 + (k – 3)^2$

$\Rightarrow k = 3 \pm 4 \Rightarrow k = 7, –1$

for k = 7 C(7,7), B(1,–1), A(4,3) are collinear.

so, $k \neq 7$

AB = BC

$\Rightarrow 9 + 16 = 36 + (k + 1)^2$

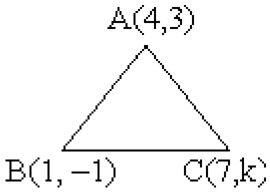
Not possible

AC = BC

$\Rightarrow 36 + (k + 1)^2 = 9 + (k – 3)^2$

$27 + k^2 + +2k + 1 = k^2 – 6k + 9$

$$\Rightarrow 8k = -19 \Rightarrow k = -\frac{19}{8}$$



Q39. If $a^2 + 9b^2 - 4c^2 = 6\ ab$ then the family of lines $ax + by + c = 0$ are concurrent at :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, - 3/2)

Ans. C, D

Sol. Given $a^2 + 9b^2 - 4c^2 = 6ab$

$\Rightarrow (a-3b)^2 = (2c)^2$

$\Rightarrow (a-3b) = \pm c$

$\therefore a - 3b + 2c = 0$ $\qquad$ $a - 3b = 2c$

$c = \dfrac{3b-a}{2}$ $\qquad$ $c = \dfrac{a-3b}{2}$

family of lines is $ax + by + c = 0$

$ax + by + \dfrac{3b-a}{2} = 0$ $\qquad$ $ax + by + \dfrac{a-3b}{2} = 0$

if these two lines are concurrent then

$3b - a = 0$

now, $ax + by + \dfrac{3b-a}{2} = 0$

$3bx + by = 0$ $\qquad$ $y = 3x$

$\therefore$ The points will be $\left(-\dfrac{1}{2}, \dfrac{3}{2}\right)$ and $\left(\dfrac{1}{2}, \dfrac{-3}{2}\right)$.

Q40. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is :

(A) isosceles (B) right angled (C) obtuse angled (D) equilateral

Ans. A, C

Sol. $x + y = 0$ ________(i)

$3x + y - 4 = 0$ ________(ii)

$x + 3y - 4 = 0$ ________(iii)

Solving lines (i) and (ii), we get

$-x = -3x + 4 \Rightarrow x = 2 \Rightarrow y = -2$

$\therefore$ (i) and (ii) intersect at A = (2, –2)

solving lines (ii) and (iii), we get

$-3x + 4 = \dfrac{4 - x}{3} \Rightarrow -9x + 12 = 4 - x \Rightarrow x = 1 \Rightarrow y = 1$

$\therefore$ (ii) and (iii) intersect at B = (1,1)

Solving lines (i) and (iii), we get

$-x = \dfrac{4 - x}{3} \Rightarrow -3x = 4 - x \Rightarrow x = -2 \Rightarrow y = 2$

$\therefore$ (i) and (iii) intersect at C = (–2,2)

So, AB, BC, AC form a triangle ABC

Now, $AB = \sqrt{(1-2)^2 + (1+2)^2} = \sqrt{10}$

$BC = \sqrt{(-2-1)^2 + (2-1)^2} = \sqrt{10}$

$AC = \sqrt{(-2-2)^2 + (2+2)^2} = \sqrt{32}$

$\therefore$ AB = BC

Since, two sides of a triangle are equal then the triangle formed by A, B, C is isosceles triangle.

Q41. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point (1, – 10). The equation of the third side can be :

(A) $x + 3y + 29 = 0$ (B) $x - 3y = 31$ (C) $3x - y = 13$ (D) $3x + y + 7 = 0$

Ans. B, D

Sol. $m_{AB} = 7$ and $m_{AC} = -1$

Let slope of BC (third side) be m.

Now $\left|\dfrac{7-m}{1+7m}\right| = \left|\dfrac{-1-m}{1-m}\right| \Rightarrow \left(\dfrac{7-m}{1+7m}\right) = \pm\left(\dfrac{-1-m}{1-m}\right)$

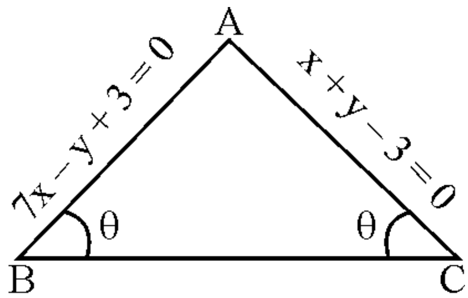
$\therefore$ Taking '+' sign, we get

$m^2 + 1 = 0$ (no real values of m possible)

And taking '–' sign, we get

$3m^2 + 8m - 3 = 0 \Rightarrow (3m - 1)(m + 3) = 0 \Rightarrow m = \dfrac{1}{3}, -3$

$\therefore$ Required equation of third side can be $x - 3y - 31 = 0$ or $3x = y + 7 = 0$



Q42. Straight lines $2x + y = 5$ and $x - 2y = 3$ intersect at the point A . Points B and C are chosen on these two lines such that AB = AC. Then the equation of a line BC passing through the point (2, 3) is :

(A) $3x - y - 3 = 0$ (B) $x + 3y - 11 = 0$ (C) $3x + y - 9 = 0$ (D) $x - 3y + 7 = 0$

Ans. A, B

Sol. Note that the lines are perpendicular. Find the equation of the lines through (2, 3) and parallel to the bisectors of the given lines, the slopes of the bisectors being – 1/3 and 3

Q43. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ. Which of the following relations hold good?

(A) $m_1 + m_2 = 5/4$ (B) $m_1m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\dfrac{2}{5\sqrt{5}}\right)$

(D) sum of the abscissa and ordinate of the point P is – 1

Ans. B, C, D

Sol. $f(x, y) = (x + 2y + 3)(3x + 4y + 5)$

$$m_1 + m_2 = -\frac{2h}{b} = -\frac{10}{8} = -\frac{5}{4}$$

$$m_1m_2 = \frac{a}{b} = \frac{3}{8} \quad \Rightarrow \quad \text{(B) is correct}$$

also L_1 and L_2 intersect at (1, – 2)

∴ sum of abscissa and ordinate = – 1 ⇒ (D) is correct

tan θ = = =

sin q = Þ q = sin–1 Þ (C) is correct

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b} = \frac{2\sqrt{25-24}}{11} = \frac{2}{11}$$

$$\sin\theta = \frac{2}{5\sqrt{5}} \qquad \Rightarrow \theta = \sin^{-1}\left(\frac{2}{5\sqrt{5}}\right) \quad \Rightarrow \quad \text{(C) is correct}$$

Q44. Let B (1, – 3) and D (0, 4) represent two vertices of rhombus ABCD in (x, y) plane, then coordinates of vertex A, if ∠BAD = 60°, can be equal to :

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

(C) $\left(\frac{1-14\sqrt{3}}{2}, \frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2}, \frac{1+2\sqrt{3}}{2}\right)$

Ans. A, B

Sol. Let inclination of AC is θ

$$\tan\theta = \frac{1}{7} \qquad \sin\theta = \frac{1}{5\sqrt{2}}, \qquad \cos\theta = \frac{7}{5\sqrt{2}}$$

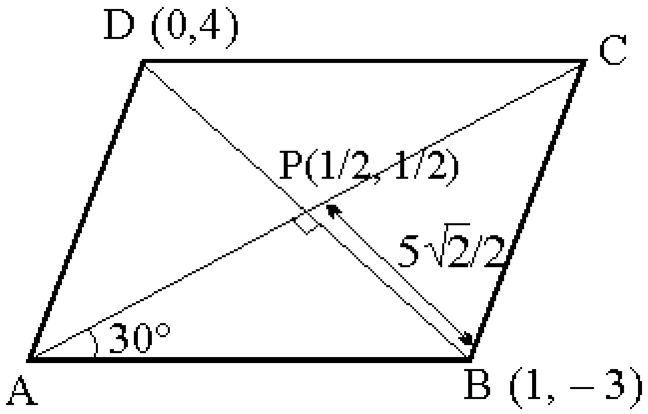
$$AP = \frac{5}{2}\sqrt{2}\cot 30^\circ = \frac{5\sqrt{2}}{2}\sqrt{3}$$

$$A = \left(\frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\,\frac{7}{5\sqrt{2}}, \frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\,\frac{1}{5\sqrt{2}}\right)$$

$$\text{or} \left(\frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\,\frac{7}{5\sqrt{2}}, \frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\,\frac{1}{5\sqrt{2}}\right)$$

$$\Rightarrow A = \left(\frac{1}{2} - \frac{7\sqrt{3}}{2}, \frac{1}{2} - \frac{\sqrt{3}}{2}\right) \quad \text{or}$$

$$\left(\frac{1}{2} + \frac{7\sqrt{3}}{2}\right), \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)$$



Q45. The equation of the bisectors of the angle between the two intersecting lines : $\frac{x-3}{\cos\theta}=\frac{y+3}{\sin\theta}$ and $=\frac{x-3}{\cos\phi}=\frac{y+5}{\sin\phi}$ are $\frac{x-3}{\cos\alpha}=\frac{y+5}{\sin\alpha}$ and $\frac{x-3}{\beta}=\frac{x+5}{\gamma}$ then

(A) $\alpha=\frac{\theta+\phi}{2}$ (B) $\beta=-\sin\alpha$ (C) $\gamma=\cos\alpha$ (D) $\beta=\sin\alpha$

Ans. A, B, C, D

Sol. -

Q46. Equation of a straight line passing through the point (4, 5) and equally inclined to the lines,
3x = 4y + 7 and 5y = 12x + 6 is

(A) 9 x – 7 y = 1 (B) 9 x + 7 y = 71 (C) 7 x + 9 y = 73 (D) 7 x – 9 y + 17 = 0

Ans. A, C

Sol. The lines will pass through (4, 5) & parallel to the bisectors between them
$\frac{3x|4y-7}{5}=\pm\frac{12x-5y+6}{13}$
by taking + sign, we get 21x + 27y + 121 = 0. Now by taking – sign, we get 99x – 77y – 61 = 0
so slopes of bisectors are $\frac{7}{9},\frac{9}{7}$
Equation of lines are $y-5=\frac{-7}{9}(x-4)$ and $y-5=\frac{9}{7}(x-4)$
⇒ 7x + 9y = 73 and 9x – 7y = 1

Q47. One side of a rectangle lies along the line 4x + 7y + 5 = 0. Two of its vertices are (–3, 1) and (1, 1). Then the equations of other sides are :

(A) 7x – 4y + 25 = 0 (B) 7x + 4y + 25 = 0 (C) 7x – 4y – 3 = 0 (D) 4x + 7y = 11

Ans. A, C, D

Sol.

Line $\perp$ to $4x + 7y + 5 = 0$ is $7x - 4y + l = 0$

It passes through (– 3, 1) and (1, 1)

$-21 - 4 + \lambda = 0 \Rightarrow \lambda = 25$

$7 - 4 + \lambda = 0 \Rightarrow \lambda = -3$

Hence lines are $7x - 4y + 25 = 0$, $7x - 4y - 3 = 0$

line $\parallel$ to $4x + 7y + 5 = 0$ passing through (1, 1) is $4x + 7y + \lambda = 0$.

Q48. Suppose ABCD is a quadrilateral such that the coordinates of A, B and C are (1, 3), (–2, 6) and (5, – 8) respectively. For what choice of coordinates of D will make ABCD a trapezium ?

(A) (3, – 6) (B) (6, –9) (C) (0, 5) (D) (3, –1)

Ans. A, B, C, D

Sol. -

Q49. The points A(0, 0), B(cosα, sinα) and C(cosβ, sinβ) are the vertices of a right angled triangle if:

(A) $\sin\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (B) $\cos\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$ (C) $\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$

(D) $\sin\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

Ans. A, B, C, D

Sol. -

Q50. Two vertices of the Δ ABC are at the points A(– 1, – 1) and B(4, 5) and the third vertex lines on the straight

line $y = 5(x - 3)$. If the area of the Δ is 19/2 then the possible co–ordinates of the vertex C are:

(A) (5, 10) (B) (3, 0) (C) (2, – 5) (D) (5, 4)

Ans. A, B

Sol. -

Q51. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x - \sqrt{3}\,y = 0$ then the co–ordinates of the third vertex are :

(A) $(0, a)$ (B) $\left(\frac{\sqrt{a}}{2}, -\frac{a}{2}\right)$ (C) $(0, -a)$ (D) $\left(-\frac{\sqrt{a}}{2}, \frac{a}{2}\right)$

Ans. A, B, C, D

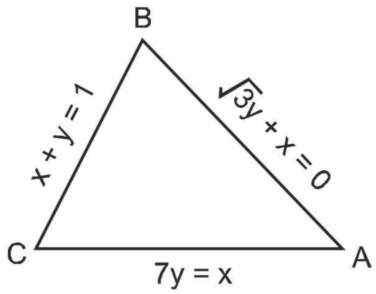
Sol. Make a figure and interpret.

Q52. The sides of a triangle are the straight lines $x + y = 1$; $7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior point of the triangle ?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

Ans. C, B

Sol.



Let slope of given lines $m_1 = \frac{1}{7}, m_2 = \frac{-1}{\sqrt{3}}, m_3 = -1$

Hence interior angle of triangle

$$tanA = \frac{m_1 - m_2}{1 + m_1m_3} = \frac{-\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{7\sqrt{3}}} = \frac{\sqrt{3}+7}{7\sqrt{3}-1} > 0$$

$$tanB = \frac{m_2 - m_3}{1 + m_2m_3} = \frac{-\frac{1}{\sqrt{3}} + 1}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} > 0$$

$$\Rightarrow\ tanC = \frac{m_3 - m_1}{1 + m_2m_1} = \frac{-1 - \frac{1}{7}}{1 - \frac{1}{7}} = \frac{-8}{6} < 0$$

Hence angle C is obtuse therefore circumcentre and orthocentre lies outside the triangle.

Q53. Match the column

Column –I	Column – II
(A) Slope of line bisecting co–ordinate axis, is	(P) 3
(B) Area of Δ formed by line $3x + 4y + 12 = 0$, with co–ordinate axis is	(Q) 1
(C) If the equation $2x^2 - 2xy - y^2 - 6x + 6y + c = 0$ represents a pair of lines then 'C' is	(R) 6
(D) If distance between the pair of parallel lines	(S) – 1

$x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $\sqrt{2}$,
then 'a/5 is equal to

(A) A – QS, B – R, C – QS, D – R (B) A – QS, B – R, C – P, D – QS

(C) A – R, B – P, C – QS, D – QS (D) A – R, B – S, C – P, D – P

Ans. 5f01555e9fff15766e370e28

Sol. -

Q54. Column–I
Column–II
(A) If a, b, c are in A. P., then lines ax + by + c = 0 are concurent at (P) (–4, –7)
(B) A point on the line x + y = 4 which lies at a unit distance from the line (Q) (–7, 11)
4x + 3y = 10 is
(C) Orthocentre of trianlge made by lines x + y = 1, x – y + 3 = 0, (R) (1, –2)
2x + y = 7 is
(D) Two vertice of a triangle are (5, –1) and (–2, 3). If orthcentre is the (S) (–1, 2)
origin then coordinated of the third vertex are (T) (0, 0)

(A) A – R, B – Q, C – S, D – P (B) A – P, B – Q, C – S, D – R

(C) A – R, B – S, C – Q, D – P (D) A – Q, B – R, C – P, D – S

Ans. 5f0157234c298c768332c388

Sol. -

Q55. Consider the lines given by
$L_1 = x + 3y - 5 = 0$
$L_2 = 3x - ky - 1 = 0$
$L_3 = 5x + 2y - 12 = 0$
Match the statements / Expression in Column–I with the statements / Expressions in Column–II and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in OMR.

Column–I	Column–II
(A) L_1, L_2, L_3 are concurrent, if	(P) $k = -9$
(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q) $k = -\frac{6}{5}$

(C) L_1, L_2, L_3 form a triangle, if (R) $k = \frac{6}{5}$

(D) L_1, L_2, L_3 do not form a triangle, if (S) $k = 5$

(A) (A) S (B) R (C) P,Q (D) P,Q,S (B) (A) P,Q (B) S (C) P,Q,S (D) R

(C) (A) S (B) P,Q (C) R (D) P,Q,S (D) (A) P,Q,S (B) P,Q (C) R (D) S

Ans. 5f015ad89fff15766e370edd

Sol. -

Q56. Let ABC be a triangle such that the coordinates of A are $(-3, 1)$. Equation of the median through B is
$2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column–I with their corresponding correct entries of column–II.

Column–I	Column–II
(A) Equation of the line AB is	(P) $2x + y - 3 = 0$
(B) Equation of the line BC is	(Q) $2x - 3y + 9 = 0$
(C) Equation of CA is	(R) $4x + 7y + 5 = 0$
	(S) $18x - y - 49 = 0$

(A) (A) R; (B) S ; (C) Q (B) (A) R; (B) Q ; (C) P (C) (A) R; (B) P ; (C) Q

(D) (A) P; (B) Q ; (C) S

Ans. 5f015b93f6c4ee7669720861

Sol. x + 3y – 5 = 0, 3x – xy – 1 = 0,5x + 2y – 12 = 0

(A) For concurrency

$$\begin{vmatrix} 1 & 3 & -5 \\ 3 & -k & -1 \\ 5 & 2 & -12 \end{vmatrix} = 0$$

$\Rightarrow$ (12k + 2) – 3(–36 + 5) – 5(6 + 5k) = 0

$\therefore$ k = 5

(B) For parallel

either $-\frac{1}{3} = \frac{3}{k}$ $\therefore$ k = –9

or $-\frac{5}{2} = \frac{3}{k}$ $\therefore k = \frac{-6}{5}$

(C) They will form triangle

when k $\neq$ 5, –9, $\frac{-6}{5}$

(D) They will not form triangle

when k = 5, –9, $\frac{-6}{5}$